RESEARCH METHODS FOR COMPUTING AND TECHNOLOGY (CT098-3-2-RMCT)
INDIVIDUAL ASSIGNMENT
PART 2 - Methodology

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Limitations of Traditional ERP Demand Forecasting in Retail SMEs: An Interpretable TKAN-Based Approach

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Abstract - Although retail Small and Medium Enterprises (SMEs) play a vital role in driving the economy of Malaysia, legacy Enterprise Resource Planning (ERP) forecasting tools often result in severe disruptions to their operations. Within the fluid e-commerce sphere, the linear model used in traditional business cannot expose non-linear consumer purchasing behavior, which keeps causing the costly Bullwhip Effect. A review of existing literature points to a critical gap, that while advanced deep learning models such as Long Short-Term Memory (LSTM) improve accuracy, its 'black box' nature inhibits managerial trust and adoption of deep learning. This research proposes integrating Temporal Kolmogorov-Arnold Network in existing ERP environments. By replacing fixed nodes with their learnable B-spline activation functions on the edges of the network, TKANs learn complex non-linear dependencies while leaving behind some of the symbolic mathematical capabilities for understanding the network. To verify this approach, this study incorporates a mixed-methods research design to target SME managers and inventory planners. A mixture of purposive and cluster sampling is used to distribute quantitative online surveys and conduct qualitative semi-structured interviews. Ultimately, this transparent layer of intelligence is attempting to mitigate forecast errors, minimize inventory holding costs and give SME managers the mathematical verifiability of precision they need to trust automated planning.

Index Terms – Bullwhip Effect, Demand Forecasting, Enterprise Resource Planning (ERP), Retail SMEs, Temporal Kolmogorov-Arnold Network (TKAN)

1.0 Introduction

Small and medium enterprises (SMEs) are a huge factor in Malaysia's national GDP, so proper inventory management is very important for economic stability. However, SMEs are finding it hard to optimise inventory turnover with legacy ERP forecasting modules that use static, linear statistical models. These traditional methods don't adapt to the highly volatile and non-linear fluctuations in demand that are so common in the modern retail, and this can often result in the Bullwhip Effect, where small fluctuations in demand lead to massive disruptions in the supply chain. While advanced deep learning models provide a greater degree of accuracy, their 'black box' nature, they cannot be trusted by SME managers, who tend to freeze out the automated recommendations resulting in manual overrides. To rectify this, a Temporal Kolmogorov-Arnold Network (TKAN) forecasting model is proposed in this research by utilizing learnable B-spline activation functions to provide a non-linear prediction with a high level of accuracy while also enabling a mathematical interpretability for accurate inventory planning activities.

In order to bridge the gap between being able to accurately predict and being able to trust the managerial, this proposal establishes the research methodology and theoretical basis for implementing the TKAN-based solution. The literature review first provides a map of the current research domain, examines the available forecasting technologies, and cites the all-important need for algorithmic transparency.

Building on such insights, the methodology identifies particular targeted user demographics and explains the rationality of a mixed-methods approach, specifying the purposive and cluster sampling approaches to the quantitative and qualitative approaches to data gathering. Finally, the conclusion presents a synthesis of such findings to show how this approach will lead to a feasible effective, trust-centric forecasting tool for SMEs in Malaysia.

2.0 Literature Review

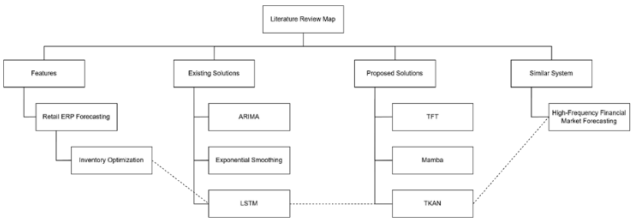

Figure 1 Literature Review Map

2.1 Features

Enterprise Resource Planning (ERP) system is considered as the backbone of the overall operations for management of business resources and supply chain stability. Within these infrastructures, demand forecasting exists as the base predictive machine that is intended to work with historic sales information and make future predictions of consumer demand. Despite their importance, traditional forecasting modules are often based on hard statistical assumptions which are not feasible in such highly nonlinear and erratic purchasing patterns typical of modern-day e-commerce markets (Chowdhury et al., 2025). Recent comparative analysis confirms the idea that univariate statistical techniques are far behind modern techniques for prediction of complex retail sales with multivariate deep neural networks (Theodoridis & Tsadiras 2025). Consequently, sudden changes in the market are often the reason for slow system response times and poor predictive accuracy. Such unacceptable functionality is a direct challenge in the process of supply chain operations and consistency of supply chain planning of the inventory distributed to the vulnerable Malaysia SMEs.

Beyond these functional constraints, conventional predictive models have the problem of being unable to improvise sudden external shocks of the market. One major weakness is their inflexibility when it comes to dynamic changes in the environment such as viral social media trends or sudden promotional shows. Saricioglu, Genevois and Cedolin (2024) point out that the heaviness of reliance on these traditional, one-step forecasting methods have the effect of systematically exaggerating demand distortion, outing forth the Bullwhip Effect (Lee et al, 1997). The knowledge requirements of modern retail inventory optimization are highly adaptive predictive frameworks capable of responding to rapidly changing consumer behavior and high intensity localized demand spikes. If the forecasting mechanisms are based on the sole fact that they map the historical data linearly, then their accuracy becomes extremely poor in the event of any disruption in the market thereby putting the financial health of the business on fundamentally invalid footing.

2.2 Existing Solutions

The viability of current demand forecasting systems is largely evaluated across three important criteria, predictive power amid non-linear market volatility, computational performance for legacy ERP integration and

algorithmic transparency that is needed for managerial confidence. Traditional statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing are computationally light, but mostly fail to cross the accuracy hurdle. To try and fill this accuracy gap, modern supply chain systems have focussed increasingly on the use of advanced machine learning models. Studies show that advanced forecasting using multi-step machine learning algorithm outperforms traditional moving average in terms of constraining the amplification of demand and improving order fulfillment rate on a big margin (Saraicioglu et al., 2024).

Among these solutions, standard Long Short-Term Memory (LSTM) networks have become a dominant baseline as they are superior to processes that can deal with sequential time-series data. However, despite the fact that the standard LSTMs models are successful in fulfilling criteria of computational efficiency and general accuracy, the basic architectures of the LSTMs have no specific interpretability. Because their outputs are based entirely on hidden neural correlations, not transparent, mathematical rationales, these 'black box' models are not good at providing actionable, verifiable insights to human decision-makers. As a result of this SME managers are not able to believe in the reasonings behind automated inventory recommendations and they engage in systematic managerial overrides, thus perpetuating Bullwhip Effect.

2.3 Proposed Solutions

In order to overcome the shortcomings of the above standard black box model methods, this is a work to adapt the Temporal Kolmogorov-Arnold Network (TKAN) framework, towards legacy ERP systems. Introduced by Genet and Inzirillo (2024), the TKAN architecture can be considered as an important paradigm shift, because it is the learnable B-spline activation functions to the edges of the network together with a direct management of temporal sequences that bring into the heart of the T-KAN neural network. Unlike pure models of deep learning, the proposed TKAN architecture include use of symbolic regression that mathematically models the internal processing of the system in readable and easily-understood formulas.

In order to know the functional superiority of TKANs, it is necessary to dig into the mathematical theorem on which the architecture has been built. Standard Multi-Layered Perceptrons (MLPs) and LSTMs are based upon fixed activation functions like ReLU, Sigmoid or Tanh which are being applied in structural nodes. In contrast, the architecture of TKAN is grounded on the so-called Kolmogorov-Arnold representation theorem and can be represented by: (Liu et al., 2024). Instead of being fixed node activations, TKANs use univariate learnable B-spline functions of the edges of the network. As the network learns about the volatility of any retail data the training happens, these B-splines changes their shapes dynamically so as to map the exact mathematical contours of demand curve. This localised learning capability subjectively returns in the form of a symbolic equation the learned demand function (Genet & Inzirillo, 2024). Consequently, this allows the system to represent a complex temporal supply chain patterns into reachable/human readable algebraic expressions.

This spline based interpretable machine learning has been very successful in the volatile forecasting domains. For example, some of the recent uses of KAN principles in energy demand forecasting have been able to capture complex sequential relationships that traditional models have been unable to detect (Zhang et al., 2025). Furthermore KAN architectures are more accurate in prediction of standard Multi-Layer Perceptron (MLPs) with lower number of trained parameters, which proves the computational efficiency that has to be incorporated into SME legacy systems (Arslan & Dönmez, 2025). By combining data prognoses with symbolic mathematical proofs, TKAN model helps human operators to confirm the credibility of organic automation in managing inventory planning in the case of a demand spike to affirm the true non-linear market mechanisms of a supply chain disruption.

2.4 Similar Systems

A recent paper by Makinde (2026) titled "Temporal Kolmogorov-Arnold Networks (T-KAN) for High-Frequency Limit Order Book Forecasting: Efficiency, Interpretability, and Alpha Decay" has shown the application of spline-based mathematical interpretability in deep learning models for forecasting complex quantitative finance models. The architecture of their proposed system consisted of the combination of learnable B-spline activation functions and temporal network capable of processing the real-time stock market trading data in an efficient way. By obtaining high frequency limit order data during periods of high volatility across the markets, the researcher was able to let the neural network map exact and non-linear mathematical relationships in the neural net's training phase. As a result, the spline-based framework dramatically improved the accuracy of making real-time predictions of how price movements would change, and would insure that the outputs would be mathematically clear, instead of being opaque statistical correlations as were usual baseline models.

Although Makinde (2026) used this TKAN architecture in the quantitative finance industry in order to keep track of high-frequency trading processes, the application here differs from past applications in using this same technological foundation to stabilize the demand of ERP forecasting for retail. The fact that the power of the TKAN network to sustain strictly predictive and interpretable qualities is becoming not diminished by pastoring severe challenges such as quickly shifting market shocks what makes the implications of the functional implementation strength during each request. By mathematically translating output from the data based on formulae from symbolic regression, the technique has been able to avoid 'the black box' effects of failing trust that are commonly displayed when using standard deep learning models. This strong performance across industries validates the integration of TKAN principles as a very viable and practical way to come up with trustworthy actionable inventory forecasts.

3.0 Methodology

This research is an experimental-quantitative approach, which includes the extraction, the processing, and the statistical evaluation of the numerical time-series data. The main goal is to benchmark the computational efficiency and the predictive accuracy of the

proposed data provider with traditional ERP Forecasting models.

3.1 Target User

In this research, the different target users are classified according to the level of interaction with the forecasting system and their position in the retail supply chain. This classification figures out the difference between those who perform the work of means of the software to make data-driven decisions and those who are the larger ecosystem of those who benefit from the stability of the operations.

3.1.1 Direct User

The direct users of the proposed TKAN based forecasting system mainly include SME managers, inventory planners and procurement officers who are responsible for daily supply chain operations. These users interact directly with the ERP interface, feeding the historical sales data and promotional calendars and local market variables the model needs to operate. Unlike traditional "black box" neural networks, these users in the TKAN architecture are given symbolic mathematical interpretations of the predicted demand and, therefore, can check the logic behind making a forecast before they actually make a purchase order. By capitalizing upon this transparent intelligence, the direct users will be able to greatly lessen the amount of manual managerial overrides, which are usually the main factor driving the Bullwhip Effect, shifting their workflow from reactive management of a crisis to proactive, evidence-based inventory planning.

3.1.2 Indirect User

Indirect users refer to the stakeholders who do not use the forecasting platform but are positive strong users of the results of its application. This category includes upstream suppliers and wholesalers, who benefit from the improved stability of orders, as well as more predictable demand patterns, that occur when SMEs use more precise forecasting tools. At the consumer level, retail customers are indirect beneficiaries who benefit from an improved product availability and fewer incidents of out-of-stock conditions for basic products. Furthermore, the SME business owners, institutional stakeholders are considered as indirect users as they benefit from the long-term financial health of the enterprise. By ensuring more efficient turnover of inventory and less holding costs, the system spreads better cash flow to these stakeholders and helps mathematicians verify its users with a verifiable tool to keep business going against the risk of volatile changes in a market.

3.2 Sampling Method

To ensure that the data gathered is both technically viable to be used in training the Temporal Kolmogorov-Arnold Network (TKAN) and representative of the world of retail in general, in this study a mixed sampling strategy is implemented, combining both the purposive and cluster sampling techniques. This dual approach is chosen as this would provide the perfect balance of stringent data quality prerequisites for more advanced machine learning on one hand and generalisability of results across varying retail segments on the other.

The first stage of the process makes use of a non-probability sampling method called purposive sampling to select subjects, so that they can be chosen consciously, based on certain features that are relevant to technical goals of the study. For the purpose of this research, these criteria are used to select retail SMEs in Malaysia with available digital ERP infrastructures and an established twenty-four month of structured historical sales records. This targeted selection is critical, as was proven by Theodoridis and Tsadiras (2025) as high quality and continuous historical data is an absolute prerequisite to successfully train multivariate deep neural networks. By having filtering for technological maturity, the research makes sure that the baseline data is robust enough to support the complicated to learn B-spline activation functions and symbolic regression TKAN architecture makes necessary.

The second stage uses the method of cluster sampling, a probability-based technique in which the population is divided into natural groups or sub-sectors. Within the scope of this article, these clusters are defined as specific retail niches within the Kuala Lumpur region in the areas of fast moving consumer goods (FMCG), consumer electronics and apparel. Rather than analyzing products, individual product SKUs, the entire clusters of sub-sector data are evaluated. This application is vitally important in ensuring the viability of the TKAN model will be rigorously stress-tested with a myriad of unique, non-linear behaviors of the market and localized demand spikes; through an efficient mechanism which will validate the system's ability to manage mitigation of the Bullwhip Effect in the broader retail supply chain (SarayCioglu et al., 2024).

3.3 Data Collection Method

This research will use a mixed data method of quantitative survey and qualitative interview to make sure a complete picture on ERP forecasting challenges and system requirements among retail SMEs is understood. The main aim of this dual approach is both to collect measurable, statistical data about current practices and systemic limitations of forecasting, and to achieve in-depth, experiential views on user trust, predictive interpretability and operational readiness for the uptake of advanced, examples of such architectures such as the Temporal Kolmogorov-Arnold Network (TKAN).

The quantitative data will be gathered, using an online survey distributed through Google Forms to the targeted demographic comprising of SME managers and inventory planners and procurement officers. This survey is a constellation of structured, 5-point Likert-scale questions aimed at empirically assessing factors such as satisfaction with legacy ERP forecasting solution, perceived predictive accuracy, frequency of manual inventory overrides and a general openness towards adopting AI-based solutions. Web-based surveys are well known as a cost-effective, scalable and valid tool for studying technology adoption behaviors in digital environments (Al Hbushi et al., 2022; Almaiah et al., 2022). A full visual

representation of the online survey form used for collecting these data is provided in the Appendix of this proposal.

In parallel, qualitative information will be collected in the form of semi-structured interviews of a purposefully selected sample of the target population. These interviews will attempt to discuss the complex, operational challenges associated with the use of traditional ERP forecasting tools, specifically targeting issues surrounding the 'black box' trust issues, associated interpretability of automated forecasting and human decision-making processes during periods of severe demand fluctuation. Semi-structured interviews are very useful for in-depth exploration of user experiences and contextual factors of the adoption of technology in real-world conditions which cannot be fully explored through structured surveys (Algassim et al., 2023).

Ultimately, the combination of quantitative surveys and qualitative interviews allows for a level of methodological triangulation that adds a great deal to the validity and reliability of the research findings (Creswell and Plano Clark, 2017). While the survey provides generalizable statistical trends across a broad sample of retail SMEs, the interviews give the kind of rich, contextual explanations that are required to understand why managerial trust in automated systems sometimes does not work. This holistic approach helps to ensure that the measurable system requirements are met together with the human-centric interpretability parameters which directly contribute to the feasibility strong and trustworthy TKAN-based forecasting solution.

4.0 Conclusion

This research methodology presents a highly structured framework in solving the critical inefficiencies of the traditional ERP demand forecasting in the Malaysian retail SME sector. The thorough literature review showed that there was a grave imbalance in the existing systems whereby these sophisticated deep learning models, although increasing the accuracy of prediction over legacy statistical models, should be seen to have the severe drawback that their opaque, 'black box' nature greatly restricts managerial trust and operational uptake. To overcome this, the scope of this proposal particularly focuses on direct SME operators, such as managers and inventory planners, because it is these people's manual inventory overruling which is the primary cause for the Bullwhip Effect which determines whether any automated system is successful or not in their eyes. By using mixed methods of data collection to combine the comprehensive quantitative survey data of ERP limitations with the rich qualitative data of user interviews, this study can assure a holistic understanding of both the technical requirements and human behavior factors. Ultimately, this focused sampling and dual-method approach will be directly used to inform the work done in creating a Temporal Kolmogorov-Arnold Network (TKAN) architecture to ensure that the final solution performs not only mathematically verifiable accuracy, but the managerial confidence needed to provide feasible and real-world adoption.

(3143 words)

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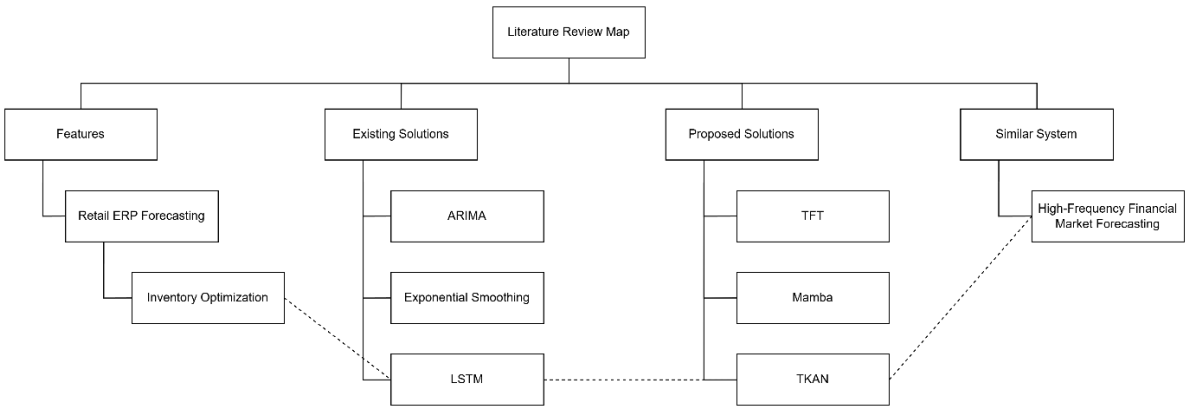
6.0 Appendix

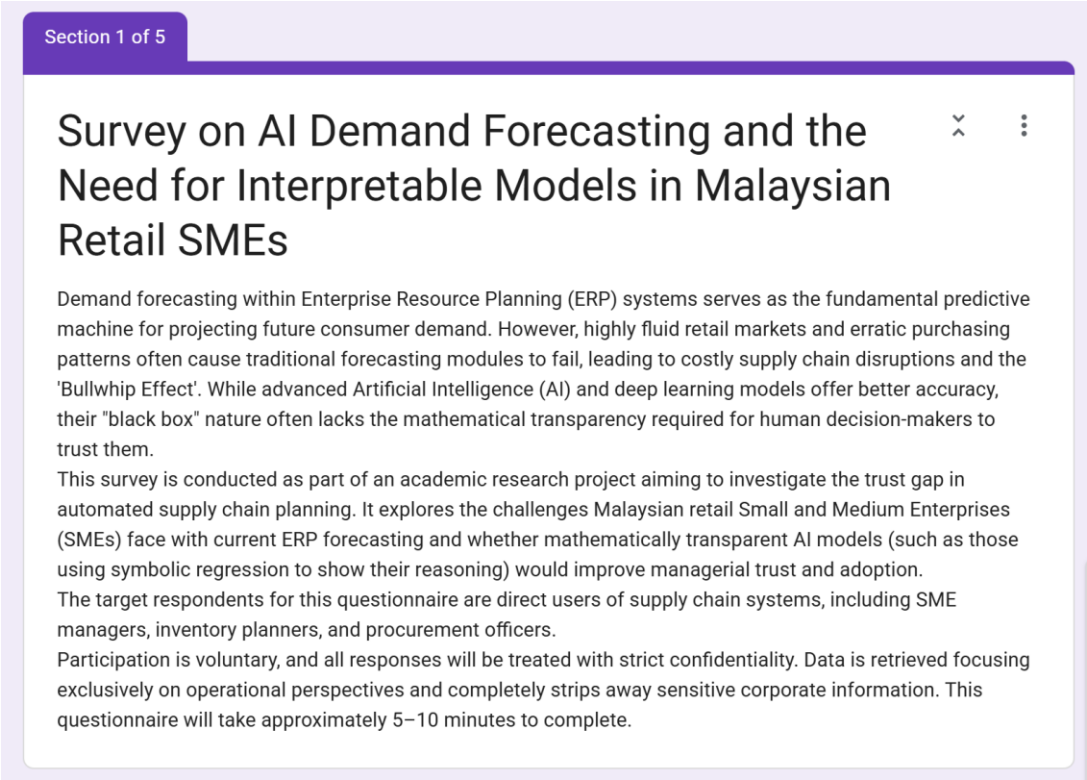
1. Literature Review Map



2. Similar System

Title	Author	Year	Technology Used	Sampling Method	Result	Limitations	Industry
T-KAN for High-Frequency Limit Order Book Forecasting	O. Makinde	2026	TKAN (B-spline activation)	High-frequency market data extraction	High accuracy & transparency over opaque models	Untested on retail/supply chain data	Finance

3. Sample Survey Questions



Section 1 of 5

Survey on AI Demand Forecasting and the Need for Interpretable Models in Malaysian Retail SMEs

Demand forecasting within Enterprise Resource Planning (ERP) systems serves as the fundamental predictive machine for projecting future consumer demand. However, highly fluid retail markets and erratic purchasing patterns often cause traditional forecasting modules to fail, leading to costly supply chain disruptions and the 'Bullwhip Effect'. While advanced Artificial Intelligence (AI) and deep learning models offer better accuracy, their "black box" nature often lacks the mathematical transparency required for human decision-makers to trust them.

This survey is conducted as part of an academic research project aiming to investigate the trust gap in automated supply chain planning. It explores the challenges Malaysian retail Small and Medium Enterprises (SMEs) face with current ERP forecasting and whether mathematically transparent AI models (such as those using symbolic regression to show their reasoning) would improve managerial trust and adoption.

The target respondents for this questionnaire are direct users of supply chain systems, including SME managers, inventory planners, and procurement officers.

Participation is voluntary, and all responses will be treated with strict confidentiality. Data is retrieved focusing exclusively on operational perspectives and completely strips away sensitive corporate information. This questionnaire will take approximately 5–10 minutes to complete.

Section 2 of 5

Professional Background & ERP Usage

This section aims to understand your role within the retail supply chain and your familiarity with inventory management systems.

Which of the following best describes your current role? *

- SME Owner / General Manager
- Inventory Planner / Controller
- Procurement / Supply Chain Officer
- IT / ERP System Administrator
- Other: _____

How many years of experience do you have in retail inventory management or supply chain operations? *

- Less than 1 year
- 1 - 3 years
- 4 - 7 years
- More than 7 years

What type of forecasting methods does your current ERP or inventory system primarily rely on? *

- Basic statistical models (e.g., Moving Averages, Exponential Smoothing, ARIMA)
- Advanced Machine Learning / AI models
- Manual calculations (e.g., Excel spreadsheets)
- I am not sure

How often do you experience sudden, unpredictable spikes or drops in consumer demand (e.g., from viral social media fads or localized promotions)? *

- Rarely
- Occasionally
- Frequently
- Very Frequently

Section 3 of 5

Current Forecasting Challenges & The Bullwhip Effect

This section explores the limitations you might face with your current forecasting tools during volatile market conditions.

My current ERP forecasting tool struggles to adapt to rapidly changing, non-linear market conditions. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

When sudden market disruptions occur, the forecasting system often provides delayed or inaccurate recommendations. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I frequently have to manually override the system's automated inventory recommendations because I do not trust its accuracy. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Small variations in consumer demand often cause exaggerated disruptions (the Bullwhip Effect) in our inventory levels and supplier orders. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Section 4 of 5

Trust & The AI "Black Box" Barrier

Advanced AI can improve accuracy, but many operate as "black boxes" where the logic behind the prediction is hidden. This section explores your trust in such systems.

I would be hesitant to trust an AI-generated purchase order if the system could not explain how it arrived at that specific number. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

The lack of transparency in how deep learning models generate recommendations makes them unsuitable for critical financial decisions. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I prefer relying on traditional statistical methods (even if they are less accurate) because their logic is easier to understand than complex AI. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

My main concern about automated AI forecasting is that I cannot verify whether the recommendation actually accounts for specific local market variables. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Section 5 of 6

Attitude Toward Interpretable AI Solutions

Imagine a new AI architecture (like a Temporal Kolmogorov-Arnold Network) that compresses complex market data into human-readable mathematical formulas, allowing you to clearly see the reasoning behind a demand spike.

Providing transparent, verifiable mathematical formulas alongside an AI forecast would significantly increase my trust in the system. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

If I could read and verify the exact variables the AI used to predict a demand spike, I would be less likely to manually override the system. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Interpretable AI models would shift my workflow from reactive crisis management to proactive, confident inventory planning. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Overall, integrating mathematical transparency into ERP forecasting modules is essential for the long-term adoption of AI in SME supply chains. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

4. Interview Questions

No.	Questions
1.	**How do you currently deal with sudden, unexpected spikes or dips in consumer demand using your existing ERP forecasting tools?**
2.	**What are the biggest limitations or frustrations you experience when relying on your current system to plan your inventory?**
3.	**Can you take me through a recent situation where you felt as though you had to manually trump the system's inventory recommendation?**
4.	**What are your personal perspectives on using advanced AI or machine learning in your daily supply chain and procurement processes?**
5.	**If someone provided you with a forecast from an AI system but the person cannot provide an explanation for how that number was calculated, how comfortable you would be using it in making major financial decisions?**
6.	**How would if you saw the exact mathematical formula or very particular variables behind AI's prediction would affect your level of trust in the software?**
7.	**When reviewing an automated purchase order recommendation, what do you need to look out for in particular to make you totally trust it without tweaking the numbers?**
8.	**If a transparent AI system was able to successfully reduce the forecasting errors and the Bullwhip Effect, how would it affect your day-to-day workload, as well as the company's overall costs for inventory?**
9.	**What areas of concern or perceived risk are the most significant as it relates to integrating a new, AI-based, forecasting architecture with your existing ERP configuration?**
10,	**Beyond accuracy and transparency, what type of specific features or capabilities would you like a new forecasting system to have in order to make your job easier?**